

Phase 2 Evaluation Results

AI / Model Evaluation and Representative User Validation
for first-generation, rural and low-income learners

Abstract

VidyaSetu is a voice-first, vernacular career-and-education bridge for first-generation, rural and low-income learners. This document reports the Phase 2 evaluation of its production MVP, covering both AI/model behaviour and representative user validation. The system was assessed through production API tests against the live deployment, local regression suites used during development, and stakeholder conversations with garment employers and a grassroots NGO in Kanpur. The AI flow passed the core MVP checks for profile extraction, goal stability, vernacular pathway generation, Sarvam voice synthesis, journey creation, proof gating and responsible-AI behaviour: 9/9 live counselor-profile personas, 5/5 vernacular pathway, journey and voice personas, 3/3 Slice5 journey-progress cases (69/69 checks) and 8/8 Slice6 smoke cases (87/87 checks), giving 156/156 journey and smoke-suite checks overall. Representative validation indicated that the operating model is legible to both sides of the bridge — employers and field partners — with the strongest guidance being the need to operate hyperlocally. We treat this as representative sprint validation rather than a large-scale field trial; the recommended next step is a documented pilot of 20–50 learner sessions.

Keywords: voice-first onboarding, vernacular AI, career and skilling guidance, proof-based journeys, Skill Passport, Sarvam TTS, Anthropic Claude, MVP evaluation.

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1 Evaluation Scope

The evaluation focused on the central promise of VidyaSetu: a learner should be able to speak or type naturally, build a usable profile without a long form, receive a realistic pathway, start a proof-based journey, and remain protected from fake or unverified opportunities.

The test set was chosen to cover the users VidyaSetu is designed for — school learners, Class 12 job seekers, low-schooling self-employment aspirants, vocational learners, informal workers, college-to-job learners, and formal skill workers. It also exercised the platform’s language and access constraints through Hinglish, Hindi, Odia, Tamil, Bengali and Marathi flows.

1.1 Why the test set is representative

The personas mirror the target segments named in the product thesis: rural or Tier-3 learners, shared-phone users, vernacular-first users, learners who need immediate income, and learners who need proof before opportunity matching. The set deliberately spans both employment and self-employment pathways, so the AI is exercised beyond a single comfortable demonstration case.

2 AI / Model Evaluation

The live MVP uses Anthropic Claude for counselor reasoning, pathway generation and journey generation, with Sarvam providing vernacular voice where available. Evaluation was run against the production URL as well as the local regression suites used during development. Table 1 reports the production validation results and Table 2 the regression-suite outcomes.

Table 1. Production validation results.

Area tested	Metric	Result	Evidence
Counselor and profile extraction	Correctly extracts goal, education, location, time, phone access, commute and proof signals from chat	9/9 passed	Live production run: <code>scripts/codex-counselor-profile-check.mjs</code>
Repeated-question regression	Replies with the next action once the profile is complete, rather than repeating the same intake question	Passed	Riya and small-business follow-up cases passed
Goal stability	Does not overwrite a specific goal with a generic study/skill path	Passed	Riya stayed customer-service/typing; mushroom stayed enterprise; mobile repair stayed mobile repair
Vernacular pathway and journey	Generates pathway and journey in the selected or spoken language style	5/5 passed	Hindi, Odia, Tamil, Bengali, Marathi
Voice readiness	Sarvam TTS returns playable audio for regional-language flows	5/5 passed	Live production run: <code>scripts/codex-vernacular-persona-check.mjs</code>
Responsible AI behaviour	Avoids guaranteed job/income claims and keeps opportunity steps behind proof/source checks	Passed	Observed across job, training and self-employment personas

Table 2. Regression-suite results.

Suite	Coverage	Result
Slice5 journey / progress / passport benchmark	Journey progress, proof readiness, passport gating and persistence	3/3 cases passed, 69/69 checks
Slice6 final smoke suite	Study, training, job, informal skill, no-location guard and proof flow	8/8 cases passed, 87/87 checks

3 Test Case Coverage

Table 3 lists the persona-level test cases and their outcomes across the six supported languages.

Table 3. Persona-level test case coverage.

Persona	Language	What was tested	Outcome
Riya — Class 12, Varanasi, computer basics, typing, customer-service job	Hinglish	Profile extraction, proof explanation, next-action reply, office-job goal stability	Passed. Meera kept the office-job goal and explained video proof in context.
Mobile repair learner — Basti, wants in-come soon	Hinglish	Vocational goal extraction and next action after profile completion	Passed. Goal stayed mobile repair and the route moved to pathway creation.
Mushroom learner — low schooling, Basti, wants self-employment	Hindi	Enterprise route; buyer/cost/scheme risk; journey and voice	Passed. Routes were mushroom-specific, not generic enterprise advice.
Poultry learner — Balangir	Odia	Self-employment path, training check, cost/buyer check and voice	Passed. Content and voice worked in the Odia flow.
Plumbing learner — Tiruchirappalli	Tamil	Local helper/training pathway and proof tasks	Passed. Route covered local helper work, practice and proof.
Tiffin learner — Nadia	Bengali	Home-food business pathway; buyer/cost/hygiene checks	Passed. Journey focused on customers, sample tiffin and hygiene.
Mobile repair learner — Nagpur	Marathi	Repair training pathway; parts/tools/safety journey and voice	Passed. Journey focused on practical repair proof.
No-location training guard	Hinglish	Whether the system avoids offline recommendations without a location	Passed in Slice6. The system asks for location instead of inventing a centre.

Main finding. The AI layer is not a static UI wrapper. It changes behaviour across job, training, study and enterprise cases; it keeps the learner goal stable; and it produces language-aware pathway and journey content with voice support.

4 Representative User Validation

During the sprint, the team validated the concept with stakeholders who would matter in a real pilot: local employers and a grassroots NGO contact in Kanpur. These conversations tested

whether the operating model made sense outside the product team. The stakeholder responses are summarised in Table 4, and the resulting learnings in Table 5.

Table 4. Representative stakeholder validation.

Stakeholder	Who was con- tacted	What was validated	Response
Employers	Clothes factories in Kanpur	Hiring through a structured, filtered learner pipeline where each candidate arrives with relevant proof, a basic profile, consent and clearer fit before screening	Positive. Employers found the concept useful because it could reduce unfiltered hiring effort and make early screening more reliable.
NGO / field partner	A representative from Shiv Shakti Foundation, Kanpur	Whether a voice-first, proof-based pathway system would fit learners who face local movement, family, phone and confidence constraints	Received with interest. The strongest advice was to make the system as hyperlocal as possible, because restrictions, safety, commute, family context and local opportunity access vary sharply by learner.

Table 5. Learnings and product implications.

Learning	Product implication
Employers are more likely to trust a candidate who arrives with proof rather than only a verbal claim.	Skill Passport and proof tasks should stay central; the product should show proof, consent and fit before any outreach.
A generic city-level path is not enough for the target audience.	The platform must capture district, block, commute limit, safe shift, phone access and local support before recommending offline steps.
Learners need a simple next step, not a broad career explanation.	Pathway pages should remain short. Meera can speak the explanation while the UI shows only route, reason and first action.
NGO/field partners need follow-up visibility.	Admin CRM, daily reminders and ADEWS-style silence alerts are important for pilot readiness.

Validation status. This was representative sprint validation, not a large-scale field trial. It is sufficient to show early demand signals and product direction. The next pilot should document 20–50 learner sessions with completion time, comprehension, trust score and follow-up outcomes.

5 Changes Driven by Evaluation

The review surfaced several concrete issues, each addressed in the MVP. These are listed in Table 6.

Table 6. Issues found during review and changes made in the MVP.

Issue found during review	Change made in MVP
The counselor repeated the same question after the learner had already answered.	Meera now gives a next-action reply — such as pressing <i>Mera Rasta / Rasta banao</i> — once enough profile data is available.
Specific goals were being overwritten by generic labels such as “skill pathway exploration.”	Goal-stability checks were added so that mushroom, mobile repair, office job and similar goals remain explicit.
Pathway pages showed too much raw content.	The UI was simplified toward short route cards, reason, pre-check and journey creation.
The learner journey was too generic.	Journey prompts now ask for skill-specific weekly tasks, free resources, proof steps and low-data instructions.
The Skill Passport could look complete before enough proof existed.	Proof/passport gating was tightened through progress and proof-readiness checks.
The product needed a way to support learners after they leave the app.	A daily WhatsApp reminder loop was added, with WATI template support and safety allowlists for live sending.

6 Risks and Remaining Work

Table 7 records the principal risks, their current handling, and the next step for each.

Table 7. Risks, current handling and next steps.

Risk	Current handling	Next step
Field usability on low-end phones	Voice, low-data UI and WhatsApp reminders are implemented.	Run a documented phone test with 3–5 actual learners on mobile data.
Over-general recommendations	Pathway and journey tests now check goal-specific route names and resources.	Add district/block-level verified centres and hyperlocal partner data.
Unsafe opportunity claims	The product avoids fake contacts and uses proof/source checks before outreach.	Keep verified employer/contact review as a mandatory worker step.
Reminder consent and message volume	WATI sending is configured with allowlists and duplicate prevention.	Add a visible consent toggle before scaling reminders beyond test users.

7 Submission Summary

VidyaSetu has a working, model-backed MVP with production validation across counselor intake, profile extraction, pathway generation, learning-journey creation, proof gating, vernacular delivery and voice. The AI evaluation passed the main automated and live checks: 9/9 counselor personas, 5/5 vernacular pathway/journey/voice personas, 3/3 Slice5 cases with 69/69 checks, and 8/8 Slice6 smoke cases with 87/87 checks.

The user-validation work showed that the concept is legible to both sides of the bridge. Kanpur garment employers responded positively to a structured, filtered hiring process with learner proof available before screening. The conversation with Shiv Shakti Foundation reinforced the need for VidyaSetu to operate hyperlocally, because learners face different constraints around commute, family permission, phone access, safety and local opportunity quality.

The strongest current evidence is the AI and product-functionality evaluation. The strongest next evidence would be a small field pilot with documented learner sessions, completion time, comprehension score, trust score, proof completion and follow-up rate.

7.1 Final assessment

VidyaSetu is ready as an MVP for a focused pilot. It demonstrates real AI functionality, vernacular voice-first onboarding, proof-based journey creation, responsible opportunity handling, admin visibility and WhatsApp reminder support. The product is not yet a scaled field deployment, but it has enough functional depth and validation evidence to justify pilot testing with NGO and employer partners.